6.4 The Triangle Midsegment Theorem

Name _____ Period ____ Date _____

Geometry 6.4 (30-minute plan) · Student sheet

I can…

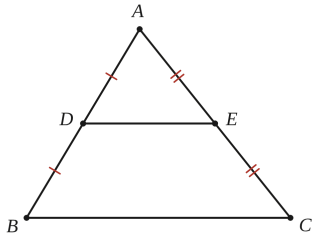
- find a midsegment of a triangle.
- use the Triangle Midsegment Theorem to find unknown lengths.

Vocabulary · fill in the missing word

Midsegment — a segment that connects the _____ of two sides of a triangle

Triangle Midsegment Theorem — the midsegment is _____ to the third side and is half as long (DE ∥ BC, DE = ½ BC)

Part 1 · Notes: fill in the blanks



Part 2 · Notes and examples (fill in each note, then work the example)

Midsegment: A **midsegment** of a triangle joins the _____ of two sides. Every triangle has _____ midsegments.

Triangle Midsegment Theorem: a midsegment is _____ to the third side and _____ as long.

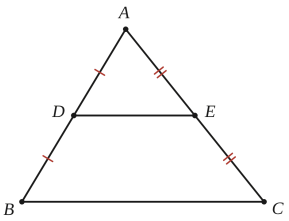
Using the theorem: If $\overline{DE}$ is a midsegment of $\triangle ABC$, then $\overline{DE} \parallel$ _____ and DE = _____ · BC.

Example 1

$\overline{DE}$ is a midsegment of $\triangle ABC$.

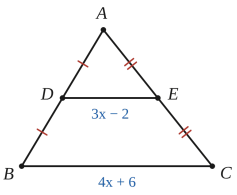
a. BC = 26. Find DE.

b. DE = 7.5. Find BC.



Example 2

DE = 3x − 2 and BC = 4x + 6. Find x and DE.



Part 3 · Partner practice

Partner practice

A(−2, 4), B(−6, −4), C(4, −2). Show the midsegment DE is parallel to BC and half as long.